Recoil transport and environmental exposure of black-hole merger remnants

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Black-hole mergers in stellar clusters can leave remnants moving through their hosts after receiving gravitational-wave recoil. Remaining gravitationally bound does not determine how soon a remnant enters a dense region or how much time it spends there. Predicting this motion requires a stellar environment that consistently supplies gravity, collisional drift, and velocity diffusion. We formulate merger-conditioned transport and define finite-region residence and species-resolved exposure on its trajectories. Public observations and published multimass-model posteriors for six Galactic clusters establish the available constraints and their shared uncertainties. For one joint realization of NGC 3201, we verify the mass distribution, potential, velocity moments, and projected tracer profiles, then evaluate anisotropic scattering from the same species distributions. A sparse black-hole component dominates the formal central diffusion, identifying a regime where the smooth closure requires a discrete-encounter test. Galactic reference orbits constructed from public phase-space coordinates supply the time-dependent external field. An analytic force bound separates a sufficient accuracy domain for isolated controls from a diagnostic tidal scale. Disk passages restrict this domain and produce rapid changes in forcing, identifying phases where the frozen-host approximation needs additional validation. These calculations establish conditional host, scattering, and environmental diagnostics; consistent host evolution and calibrated remnant residence distributions remain unresolved.

I. INTRODUCTION

A black hole formed by a merger can receive a recoil from anisotropic gravitational-wave emission. Within a stellar cluster, its subsequent motion reflects both that recoil and the state of the system in which the merger occurs. The progenitor's center-of-mass motion, merger position, host potential, and gravitational scattering determine the remnant's evolving position and velocity. A gravitationally bound remnant need not remain near the center, and entry into a dense region does not imply that the remnant comes to rest there.

The distinction matters whenever the surrounding population influences subsequent evolution. Residence measures the time spent in a specified region, whereas density exposure weights the trajectory by the material encountered along it. Neither is fixed by a retention flag alone. Their calculation requires a dynamical description with enough information to distinguish repeated excursions, continued wandering, and changes in the environment during the remnant's motion.

The necessary kinetic ingredients are established. Rosenbluth collision integrals relate velocity drift and diffusion to a field-particle distribution [1]; their gravitational application permits anisotropic velocity distributions [2]. Direct calculations of kicked black holes illustrate limitations of simplified dynamical-friction descriptions and show that individual encounters can increase a remnant's energy [3]. These results motivate treating mean drift and fluctuations together, while testing the regime in which a local diffusion closure is adequate. They do not supply a calibrated prescription for an arbitrary cluster.

We formulate two linked problems. Step 1 defines merger-conditioned transport in position and velocity, with an explicit matter inventory, reference frame, collision closure, and termination rule. Step 2 defines finite-region entry, residence, and species-resolved environmental exposure on the resulting histories. Their common output is a timed description of the remnant's occupation of its environment. The immediate question is which launch and host information must be retained to predict these observables at a declared accuracy.

We connect this formulation to an analysis of six observed Galactic clusters. Proper-motion profiles, structural estimates, and published multimass-model posteriors permit concrete comparisons of host scales and parameter dependence. Distribution-function families can supply candidate consistent hosts [4], but measured projected kinematics are not themselves local phase-space distributions. We use this observational connection to reconstruct one published joint realization for NGC 3201, verify its internal consistency, and evaluate local drift and diffusion from its mass-resolved distributions. This calculation supplies a controlled test of the host-to-collision connection, rather than a new fit or a trajectory prediction. Its species decomposition identifies where the smooth weak-scattering approximation needs particular scrutiny.

II. OBSERVATIONAL INPUTS AND COMPARISON METHOD

A. Sample and source distinctions

We examine NGC 0104 (47 Tuc), NGC 1851, NGC 3201, NGC 5139 (ω Centauri), NGC 6121 (M4), and NGC 6397. This deliberately selected sample provides varied mass and size scales and public kinematic and model products. It is not a complete cluster census or a selection-corrected population of black-hole-merger hosts.

For five clusters we use the original binned Hubble Space Telescope (HST) proper-motion products of Libralato et al. [5]. For ω Centauri we use the logarithmic radial bins released with Häberle et al. [6]. The former dispersion profiles use bright subgiant and red-giant tracers, whereas the latter use the full magnitude range. Stellar mass and selection can

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affect dispersion in a partially relaxed system; converting the units cannot remove this distinction.

We inspect the original Gaia EDR3 field catalogs and fitted radial curves of Vasiliev and Baumgardt [7], separately from compiled kinematic profiles. Membership probabilities, quality flags, measurement errors, and source identifiers describe the field catalogs. They do not make every listed star a cluster member, and raw proper-motion spread is not an intrinsic velocity dispersion. The released smooth radial curves are correlated fit products; their grid points are not independent observational bins.

Structural comparison values are taken from one captured path of the Baumgardt cluster database associated with the modeling and distance studies in Refs. [8, 9].[1] These values are model-derived estimates. In particular, a three-dimensional half-mass radius is distinct from a projected half-light or core radius. The compiled surface profiles mix normalized stellar counts and, for some sources, surface brightness; they are not directly a three-dimensional mass density.

For a separate view of host uncertainty, we inspect the released standard-fit multimass posterior samples discussed by Dickson et al. [10, 11]. Original observations, compilations, and these posterior products share underlying evidence. We keep the views separate rather than multiply them together as independent likelihood factors. The complete data collection comprises 51 captured files, including supporting source definitions and historical candidate observation bundles.

B. Native measurements, units, and shared uncertainty

For angular radius α, distance D, and proper-motion dispersion σ_μ, the small-angle conversions are

$$\frac{R}{\mathrm{pc}} = \frac{1000}{206264.806}\,\frac{D}{\mathrm{kpc}}\,\frac{\alpha}{\mathrm{arcsec}}, \tag{1}$$

$$\frac{\sigma_\perp}{\mathrm{km\,s^{-1}}} = 4.74047046\,\frac{D}{\mathrm{kpc}}\,\frac{\sigma_\mu}{\mathrm{mas\,yr^{-1}}}. \tag{2}$$

The radial and tangential proper-motion components are projected components, not radial and tangential velocities in three-dimensional space. Line-of-sight dispersions already expressed in physical velocity units require no proper-motion conversion. Native measurements and asymmetric errors are retained alongside the display conversions.

For a vector of converted quantities $\mathbf{y}(D)$, a shared distance error contributes, to first order,

$$C_{ab}^{(D)} = \frac{\partial y_a}{\partial D}\,\frac{\partial y_b}{\partial D}\,\sigma_D^2. \tag{3}$$

This is one covariance contribution, not the complete uncertainty model. Independence of distance and the measured quantities is not guaranteed, particularly for a kinematic distance. Joint propagation must retain any cross-covariances. Our plots condition on a catalog distance and show converted measurement errors only; distance derivatives and uncertainty are recorded separately. They are not added as independent errors in each radial bin.

Using total mass M and three-dimensional half-mass radius r_h from the same catalog view, define comparison scales

$$r_0 = r_h, \qquad v_0 = \sqrt{\frac{GM}{r_h}}, \qquad t_0 = \frac{r_h}{v_0}. \tag{4}$$

We use $G = 4.30091727 \times 10^{-3}\,\mathrm{pc\,(km\,s^{-1})^2}\,M_\odot^{-1}$ and convert $1\,\mathrm{pc}/(\mathrm{km\,s^{-1}}) = 0.97779222\,\mathrm{Myr}$. The quantities R/r_h and $\sigma_\perp/v_0$ provide common comparison coordinates. This does not impose a shared density profile or make the clusters homologous. Neither v_0 nor t_0 is a measured dispersion, escape speed, or relaxation time. Uncertainty in the joint mass, radius, and distance scales is not propagated into these display plots.

C. Joint host parameters

The inspected posterior products share the coordinate pattern

$$\begin{aligned}\boldsymbol{\theta}_c = (&W_0, M, r_h, \ell_a, g_\mathrm{t}, \delta, s^2, F,\\ &\alpha_1, \alpha_2, \alpha_3, f_\mathrm{ret}, D)_c,\end{aligned} \tag{5}$$

for cluster c. Here ℓ_a is the base-10 logarithm of the dimensionless anisotropy radius, g_t controls truncation, and δ governs the mass dependence of the velocity scale. These quantities belong to the specified lowered-isothermal and stellar-evolution model [4, 10]. The mass-function slopes α_i and black-hole retention parameter f_ret retain that model's conventions. The stored retention parameter is a percentage, not the present-day black-hole mass fraction. The coordinates s^2 and F are likelihood nuisance parameters rather than forces or velocity factors. The symbol g_t is separate from the geometric region function and Rosenbluth potentials below.

We normalize each stored log weight ℓ_k^w as

$$w_k = \frac{\exp(\ell_k^w - \ell_\mathrm{max}^w)}{\sum_l \exp(\ell_l^w - \ell_\mathrm{max}^w)}. \tag{6}$$

Moments and marginal quantiles are computed directly from complete weighted rows, without resampling or evaluating a host model. For example,

$$\overline{\boldsymbol{\theta}} = \sum_k w_k \boldsymbol{\theta}_k, \tag{7}$$

$$\mathsf{C}_\theta = \sum_k w_k (\boldsymbol{\theta}_k - \overline{\boldsymbol{\theta}})(\boldsymbol{\theta}_k - \overline{\boldsymbol{\theta}})^\mathsf{T}. \tag{8}$$

This is the covariance of the weighted empirical posterior; it is not a finite-sample unbiased estimator. Marginal quantiles use interpolation in the sorted weighted cumulative distribution. Nuisance coordinates remain in the joint records,

[1] The selected structural table is the database's /newdata/ parameter table. The root-path table is retained as a separate comparison; values from the two paths are not merged.

TABLE I. Source coverage. Gaia field rows include nonmembers. HST bins refer to the original proper-motion profiles; the combined, radial and tangential quantities share stars. Flagged rows belong to the separate 418-row compilation and are not corrected or used as likelihood inputs.

Cluster	Gaia field rows	HST bins	Flagged rows
47 Tuc	144,761	28	0
NGC 1851	12,688	20	4
NGC 3201	63,171	6	0
ω Cen	228,055	40	1
M4	69,417	6	0
NGC 6397	103,786	7	15
Total	621,878	107	20

and marginal summaries do not define independent parameter draws.

Exact observation-file and software-revision identifiers were not found in the inspected run metadata. Historical candidate inputs are therefore documented without claiming a reproduced fit. Current resources cannot automatically replace the inputs to an older posterior. The distinct NGC 6397 no-black-hole alternative discussed in Ref. [11] has a released summary and figures but no corresponding chain in the inspected release.

III. OBSERVATIONAL FINDINGS

A. Coverage and source reconciliation

Table I summarizes the inspected products. The six original Gaia field catalogs contain 621,878 rows, with no repeated source identifier within a cluster. The HST products provide 107 radial bins and 321 component records. Combined, projected radial, and projected tangential dispersions in a bin share stars; the combined value is not a third independent measurement. The released Gaia curves contain 101 grid points per cluster, or 606 correlated samples in total.

The separate compilation contains 418 kinematic rows, of which 20 require reconciliation: four zero-radius entries for NGC 1851, one all-zero entry for ω Centauri, and 15 NGC 6397 entries whose dispersion equals both quoted errors. Four NGC 1851 records also share one radius, source, and component key. These observations motivate flags, not guessed corrections or a claim that every flagged value is physically impossible. The compilation is retained for comparison and is not treated as an approved likelihood input. The 422 surface-profile records likewise retain their unresolved absolute mass normalization.

Figure 1 shows the variation in radial sampling. At the catalog point scales, the intersection of the six HST median-radius spans is approximately 0.081–0.251 in projected R/r_h. This is not exact shared annular coverage: most HST bin edges are unavailable, and a span does not establish sampling density between its endpoints. No common radial interpolation or likelihood is constructed.

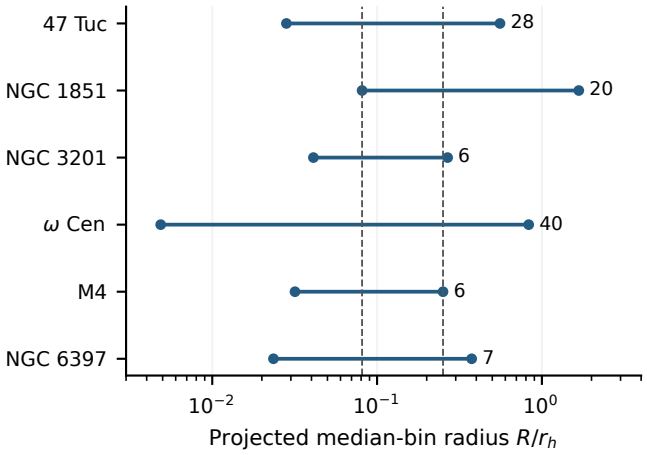


FIG. 1. Spans of original HST median-bin radii in common comparison coordinates. Numbers at the right give radial-bin counts. Dashed lines mark the intersection of the six spans at fixed catalog scales. They do not represent bin edges or a common three-dimensional domain.

B. Structural scales and projected kinematics

Table II reports the selected structural estimates and derived scales. Catalog mass ranges from $8.88 \times 10^4\,M_\odot$ for M4 to $3.32 \times 10^6\,M_\odot$ for ω Centauri; the listed central escape speeds range from 18.6 to 61.9 $\mathrm{km\,s^{-1}}$. These are conditional model estimates, not a measured retention law for merger remnants. Retention would additionally depend on position, inherited progenitor motion, recoil direction, external gravity, and the criterion used to classify escape.

The derived time scales span approximately 0.155–0.851 Myr. They establish dimensional comparison units, not return or damping times. Radius and escape-speed errors are not supplied with these catalog entries; the quoted mass and distance errors also do not capture every model systematic. Mixing a catalog mass with a radius or distance from a different fit would create an unsupported joint parameter combination.

Figure 2 compares original HST radial and tangential dispersions at the catalog scales. It displays the available projected constraints without fitting a common profile. Tracer selection, scale uncertainty, calibration, and radial sampling differ across clusters. Consequently the figure neither demonstrates universal kinematics nor statistically rejects them. HST transformations can remove sky-plane streaming signals [5]; absence of such a signal in these dispersion products is not evidence of zero cluster rotation.

C. Posterior dependence and model sensitivity

Table III reports selected marginal summaries of the published standard-fit samples. The effective sample sizes, calculated as $1/\sum_k w_k^2$, reproduce all six printed source values and range from approximately 5,171 to 6,889. Twelve param-

TABLE II. Structural-catalog estimates and derived display scales. The selected catalog path supplies the quoted mass and distance errors, but no errors for r_h or $v_{\mathrm{esc},0}$; missing uncertainty is not zero. Model systematics and joint scale uncertainty are not included. The gravitational scales v_0 and t_0 are defined in Eq. (4); t_0 is not a friction time.

Cluster	D (kpc)	$M/(10^5 M_\odot)$	r_h (pc)	$v_{\mathrm{esc},0}$ (km s^{-1})	v_0 (km s^{-1})	t_0 (Myr)
47 Tuc	4.52 ± 0.03	8.900 ± 0.050	6.24	51.0	24.77	0.246
NGC 1851	11.95 ± 0.13	2.780 ± 0.040	3.11	42.7	19.61	0.155
NGC 3201	4.74 ± 0.04	1.800 ± 0.030	8.37	18.6	9.62	0.851
ω Cen	5.43 ± 0.05	33.200 ± 0.100	9.85	61.9	38.07	0.253
M4	1.85 ± 0.02	0.888 ± 0.011	3.81	19.5	10.01	0.372
NGC 6397	2.48 ± 0.02	0.901 ± 0.009	3.17	23.4	11.06	0.280

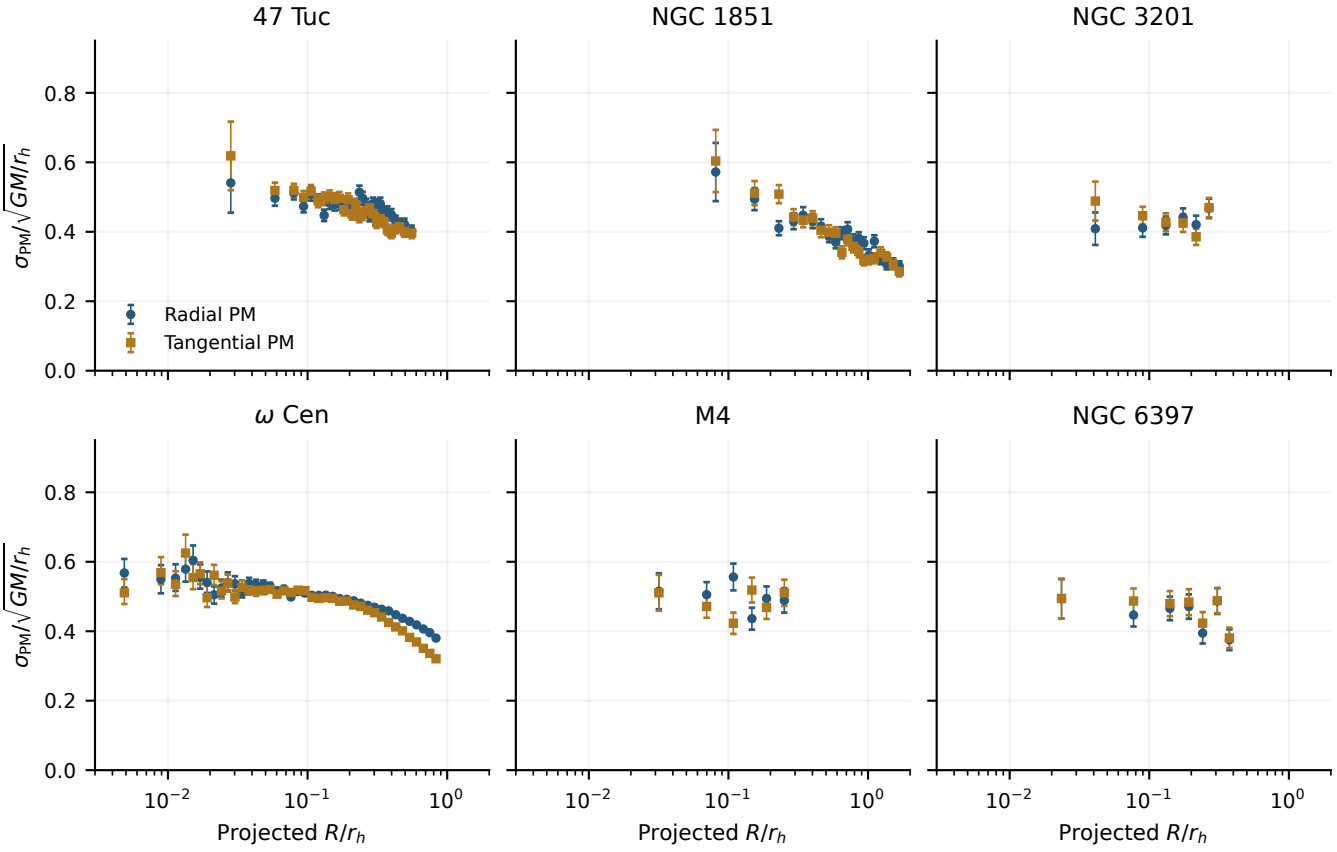


FIG. 2. Original projected radial and tangential HST proper-motion dispersions in common dimensionless coordinates. Circles and squares distinguish components. Five clusters use the bright-tracer profiles of Libralato et al. [5]; ω Centauri uses the full-magnitude-range profiles of Häberle et al. [6]. Error bars retain the source measurement-error convention and condition on catalog point scales. Shared distance and joint structural uncertainty are not fully propagated. Components and bins need not be statistically independent.

eter rows differ beyond three-decimal rounding in a mean or standard deviation. The original summary procedure averages statistics from resampled runs, whereas Eqs. (6)–(8) describe the original weighted sample. This estimator distinction permits close agreement without rounding-identical moments; it does not establish exact source-fit reproduction.

Mass and radius have weighted correlations of 0.900 in NGC 3201 and 0.944 in ω Centauri. Independent sampling of their marginal distributions would discard these dependencies. Because the scales in Eq. (4) involve ratios and products of these parameters, uncertainty should be propagated from complete host realizations. This requirement also applies to quantities derived from a realization, including its potential and species distributions.

Figure 3 compares the published multimass marginal estimates with the structural catalog. For example, NGC 6397 has catalog $r_h = 3.17$ pc, compared with a standard-fit posterior median 4.887 pc and marginal 16th–84th interval [4.747, 5.044] pc. The latter is not the unavailable alternative no-black-hole chain. This difference identifies sensitivity to

TABLE III. Direct weighted marginal medians and 16th–84th percentiles from the published standard-fit joint samples of Dickson et al. [10, 11]. These are conditional model estimates, not new fits. The NGC 6397 row is the standard fit; its distinct no-black-hole alternative lacks a released chain in the inspected product. Correlations refer to the complete weighted samples.

Cluster	$M/(10^5 M_\odot)$	r_h (pc)	D (kpc)	Corr(M, r_h)
47 Tuc	9.045 [8.996, 9.094]	6.714 [6.679, 6.749]	4.420 [4.408, 4.431]	0.327
NGC 1851	3.250 [3.211, 3.288]	3.712 [3.670, 3.757]	12.183 [12.106, 12.268]	-0.235
NGC 3201	1.757 [1.699, 1.814]	9.598 [9.070, 10.152]	4.636 [4.603, 4.671]	0.900
ω Cen	30.824 [30.693, 31.489]	9.419 [9.402, 9.529]	5.300 [5.287, 5.327]	0.944
M4	0.911 [0.894, 0.929]	3.920 [3.859, 3.982]	1.852 [1.838, 1.866]	0.316
NGC 6397	1.084 [1.061, 1.109]	4.887 [4.747, 5.044]	2.436 [2.422, 2.452]	0.782

model and data choices; it is neither temporal expansion measured between releases nor a statistical rejection using independent datasets.

D. What the comparison establishes

The analysis supports a common field and unit convention with a distinct joint host distribution for each cluster. It does not establish six interchangeable scattering backgrounds. NGC 3201 and 47 Tuc provide initial candidates for reproducing a published host and checking its projected observables, with rotation explicitly assessed for 47 Tuc. NGC 1851 and M4 extend the structural comparison. NGC 6397 tests sensitivity in a core-collapsed system, while ω Centauri requires particular attention to rotation, flattening, and mass-dependent tracer selection [6, 11].

Published density and cumulative-mass bands do not by themselves provide mutually consistent local host realizations. Independently assembling median profiles could violate the consistency between mass density, gravity, and velocity structure. These observational findings constrain how the following transport problem should be supplied with inputs. Section V evaluates a specific reconstructed realization, while preserving the distinction between numerical consistency and physical calibration.

IV. STEP 1: RECOIL TRANSPORT

A. Matter distributions and reference frame

Use physical units and cluster-relative coordinates in translating, nonrotating axes. Let $\mathbf{R}(t)$ be the reference position in an inertial frame, and let $(\mathbf{r}, \mathbf{v})$ denote the remnant's relative position and velocity. The host description Θ contains its matter distributions, gravitational boundary conditions, and any supplied temporal evolution.

For species j with individual mass m_j, define a number phase-space distribution $F_j(\mathbf{r}, \mathbf{u}, t \mid \Theta)$. Thus $F_j\, d^3r\, d^3u$ is an expected object count. Its dimensions are inverse volume times inverse velocity cubed. The number and mass densities are

$$n_j(\mathbf{r}, t) = \int F_j(\mathbf{r}, \mathbf{u}, t)\, d^3u, \tag{9}$$

$$\rho_j(\mathbf{r}, t) = m_j n_j(\mathbf{r}, t), \qquad \rho_c = \sum_j \rho_j. \tag{10}$$

The cluster potential satisfies

$$\nabla^2 \Phi_c = 4\pi G \rho_c, \tag{11}$$

together with specified boundary conditions. The tracked remnant's self-field is excluded. An unresolved binary can be represented by its center of mass only on encounter scales supporting that approximation; its mass must not also enter through separately counted components. A continuous mass spectrum requires a mass integral or a declared binning convention.

The same distributions determine streaming and velocity dispersion:

$$\overline{\mathbf{u}}_j = \frac{1}{n_j} \int \mathbf{u} F_j\, d^3u, \tag{12}$$

$$S_{j,ik} = \frac{1}{n_j} \int (u_i - \overline{u}_{j,i})(u_k - \overline{u}_{j,k}) F_j\, d^3u. \tag{13}$$

These moments are defined where $n_j > 0$. Distinct species may have different concentrations, streaming motions, and anisotropies. Density and a scalar dispersion do not generally determine the full velocity distribution needed for a collision calculation.

For an external gravitational potential $\Phi_{\rm ext}$, the relative external acceleration is

$$\mathbf{a}_{\rm ext,rel} = -\nabla \Phi_{\rm ext}(\mathbf{R} + \mathbf{r}, t) - \ddot{\mathbf{R}}(t). \tag{14}$$

If the reference orbit follows $\ddot{\mathbf{R}} = -\nabla \Phi_{\rm ext}(\mathbf{R}, t)$, Eq. (14) is the differential tidal acceleration. There are no Coriolis terms in these nonrotating axes. Internal stellar rotation remains possible through F_j.

This is a test-remnant description in a supplied background. Poisson's equation does not evolve the host. A frozen host needs a validity interval; an evolving host needs mutually consistent distributions and potential at each epoch. Backreaction, collective response, and energy feedback require additional equations and are not obtained by introducing a time argument alone.

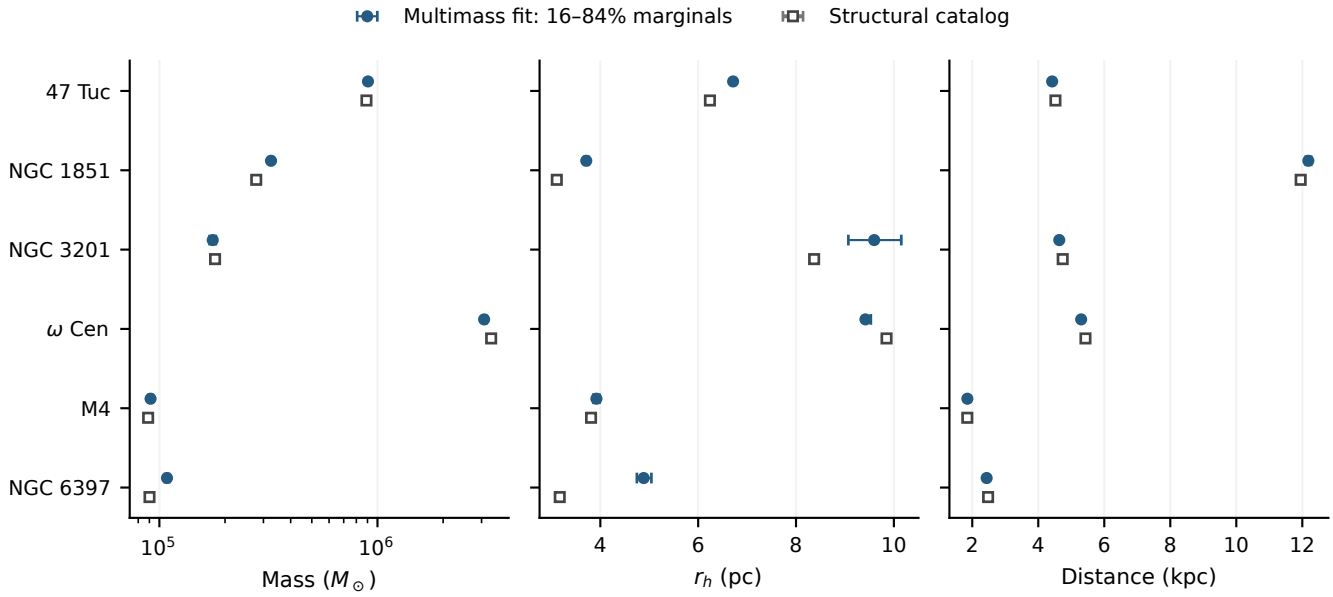


FIG. 3. Structural-catalog estimates (open squares) and published standard-fit posterior marginals (circles with 16th–84th intervals). The mass axis is logarithmic. Catalog mass and distance errors are shown where provided; catalog radius errors are unavailable. Source observations overlap, and model assumptions differ. The NGC 6397 standard fit is kept distinct from its no-black-hole alternative. The comparison measures dependence on the adopted inference, not cluster evolution.

B. Merger-conditioned initial conditions

Set elapsed time zero at the remnant-producing merger. The launch conditions are

$$\mathbf{r}(0) = \mathbf{r}_{\rm m}, \qquad \mathbf{v}(0) = \mathbf{v}_{\rm COM} + \mathbf{v}_{\rm k}, \tag{15}$$

$$m_{\rm r}(0) = m_{\rm rem}. \tag{16}$$

Here $\mathbf{v}_{\rm COM}$ is the progenitor binary's center-of-mass velocity, not its internal orbital velocity. The recoil vector $\mathbf{v}_{\rm k}$ must be transformed into the cluster frame before addition. Equation (15) uses nonrelativistic cluster kinematics; a relativistic merger calculation supplies recoil and remnant mass.

Let $\mathcal{D}$ denote observational or simulation constraints and $\mathcal{M}$ the merger-selection rule and relevant assembly information. The initial law is joint:

$$z_0 = (\mathbf{r}_{\rm m}, \mathbf{v}_{\rm COM}, \mathbf{v}_{\rm k}, m_{\rm rem}), \tag{17}$$

$$z_0 \sim \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}). \tag{18}$$

No independence among these variables or from the host is imposed. In particular, the initial specific angular momentum $\mathbf{r}_{\rm m}\times(\mathbf{v}_{\rm COM}+\mathbf{v}_{\rm k})$ need not vanish. A noncentral trajectory need not pass through the exact center.

A retained-only population is meaningful only after specifying the escape criterion and the interval over which retention is assessed. Initial negative cluster-relative energy does not by itself establish permanent retention in an evolving tidal environment. The base transport process is defined before any selection on future survival. Statistics conditioned on that survival require conditioning or reweighting its full path law; changing the launch distribution alone while leaving subsequent stochastic evolution unchanged is insufficient.

C. Velocity drift and diffusion

Define the smooth acceleration $\mathbf{a}_g = -\nabla\Phi_{\rm c} + \mathbf{a}_{\rm ext,rel}$. A local diffusion description takes the Itô form

$$d\mathbf{r} = \mathbf{v}\,dt, \tag{19}$$

$$d\mathbf{v} = (\mathbf{a}_g + \mathbf{A})\,dt + \mathbf{B}\,d\mathbf{W}, \tag{20}$$

$$\mathbf{D} = \mathbf{B}\mathbf{B}^{\mathsf{T}}. \tag{21}$$

The three Wiener components satisfy $\mathbb{E}[dW_i\,dW_k] = \delta_{ik}\,dt$. After subtracting smooth gravity, the first and second collisional increments define

$$A_i = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll} \mid \mathbf{r}, \mathbf{v}, t]}{\Delta t}, \tag{22}$$

$$D_{ik} = \lim_{\Delta t\to 0} \frac{\mathbb{E}[\Delta v_i^{\rm coll}\Delta v_k^{\rm coll} \mid \mathbf{r}, \mathbf{v}, t]}{\Delta t}. \tag{23}$$

These limits belong to a coarse-grained approximation; they do not resolve the duration of each gravitational encounter. The dimensions of $\mathbf{A}$ and $\mathbf{D}$ are acceleration and velocity squared per time, respectively. Once $\mathbf{A}$ is the complete Itô first moment, an additional noise-induced drift must not be appended.

Between discrete transitions, remnant mass is fixed on each trajectory. At fixed $\mathcal{Q} = (m_{\rm r}, \Theta, \mathcal{D}, \mathcal{M})$, let $p(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q})$ denote its phase-space density. The interior evolution is

$$\partial_t p = -\partial_{r_i}(v_i p) - \partial_{v_i}[(a_{g,i} + A_i)p] + \frac{1}{2}\partial_{v_i}\partial_{v_k}(D_{ik}p). \tag{24}$$

Repeated Cartesian indices are summed. The factor of one half follows from Eq. (21). Initial, boundary, and termination conditions are needed to define a solution. When launch masses vary, each mass-dependent operator must retain its own conditioning; replacing it by an operator evaluated at a mean mass is not generally equivalent.

D. A candidate weak-scattering closure

For local, uncorrelated weak deflections, a leading-logarithm Rosenbluth closure supplies both coefficients from the same matter distribution [1, 2]. At fixed position and time, define

$$h_j(\mathbf{v}) = \int \frac{F_j(\mathbf{r}, \mathbf{u}, t)}{|\mathbf{v} - \mathbf{u}|}\, d^3u, \tag{25}$$

$$g_j(\mathbf{v}) = \int F_j(\mathbf{r}, \mathbf{u}, t)|\mathbf{v} - \mathbf{u}|\, d^3u. \tag{26}$$

For the number normalization in Eq. (9),

$$A_i = 4\pi G^2 \sum_j m_j(m_\mathrm{r} + m_j) \ln\Lambda_j\, \partial_{v_i} h_j, \tag{27}$$

$$D_{ik} = 4\pi G^2 \sum_j m_j^2 \ln\Lambda_j\, \partial_{v_i}\partial_{v_k} g_j. \tag{28}$$

A mass-normalized distribution instead absorbs a factor of m_j into each Rosenbluth potential. Mixing those conventions would change the collision coefficients.

The Coulomb logarithm $\ln\Lambda_j = \ln(b_{\max,j}/b_{\min,j})$ represents a declared range of encounter impact parameters. It is held fixed during the velocity integrals and differentiations above. Arbitrary velocity-dependent cutoffs require a consistent derivation rather than an unqualified substitution. Neither cutoff values nor a prescription are selected here.

Writing $\mathbf{w} = \mathbf{v} - \mathbf{u}$, the derivative kernels are

$$\partial_{v_i} h_j = -\int F_j \frac{w_i}{w^3}\, d^3u, \tag{29}$$

$$\partial_{v_i}\partial_{v_k} g_j = \int F_j \left(\frac{\delta_{ik}}{w} - \frac{w_i w_k}{w^3}\right) d^3u. \tag{30}$$

For nonnegative distributions and positive Coulomb logarithms, $\mathbf{D}$ is symmetric and positive semidefinite. Streaming and anisotropy enter through F_j; a Maxwellian is not imposed. Mean collisional drift need not oppose the cluster-frame velocity in every anisotropic or streaming population.

The closure excludes strong encounters and collective or long-range correlations. Its applicability depends on the scales and populations contributing to relaxation. Kicked-black-hole calculations showing departures from simplified friction models [3] therefore motivate independent validation, not an assumption that adding these coefficients is sufficient.

E. Energy, termination, and probability accounting

For fixed remnant mass, define specific cluster-relative energy

$$\epsilon = \frac{1}{2}|\mathbf{v}|^2 + \Phi_\mathrm{c}(\mathbf{r}, t). \tag{31}$$

Itô's rule gives

$$d\epsilon = \left[\partial_t\Phi_\mathrm{c} + \mathbf{v}\cdot\mathbf{a}_\mathrm{ext,rel} + \mathbf{v}\cdot\mathbf{A} + \frac{1}{2}\,\mathrm{tr}\,\mathbf{D}\right] dt + \mathbf{v}^\mathsf{T}\mathbf{B}\, d\mathbf{W}. \tag{32}$$

The diffusion contribution is required. Individual trajectories can gain energy, and this energy is not a conserved tidal escape criterion. No condition resets position or velocity upon core entry.

For a common physical horizon T, define the modeled endpoint

$$T_* = \min\{T, \tau_\mathrm{esc}, \tau_\mathrm{strong}, \tau_\mathrm{valid}\}. \tag{33}$$

The stopping times represent declared escape, a strong encounter requiring an unavailable transition model, and loss of validity of the supplied description. A strong-encounter stop needs a detector, resolved objects, or a justified event model; missing encounter information is not evidence of no encounters. Weak scattering and explicitly resolved interactions must partition deflections without double-counting.

Let $p_\mathrm{alive}(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q})$ include only paths with $t < T_*$. Then

$$\int p_\mathrm{alive}\, d^3r\, d^3v = S(t \mid \mathcal{Q}) = \Pr(t < T_* \mid \mathcal{Q}). \tag{34}$$

Survival here means continued inclusion in the transport description. It does not mean that the black hole ceases to exist when a path is stopped. The transport operator requires the corresponding absorbing boundary or justified killing mechanism. A history-dependent stop requires an augmented state or a justified history-dependent treatment; it does not automatically close Eq. (24) in position and velocity.

The density in Eq. (34) is conditional on fixed remnant mass. A joint mass density is obtained by multiplication with the initial mass marginal π_m:

$$p^\mathrm{joint}_\mathrm{alive}(\mathbf{r}, \mathbf{v}, m, t \mid \Theta, \mathcal{D}, \mathcal{M}) = \pi_m(m \mid \Theta, \mathcal{D}, \mathcal{M}) p_\mathrm{alive}(\mathbf{r}, \mathbf{v}, t \mid \mathcal{Q}_m), \tag{35}$$

where $\mathcal{Q}_m = (m, \Theta, \mathcal{D}, \mathcal{M})$. Integrating a fixed-mass conditional density over mass without this weight would not represent the launch ensemble.

V. RECONSTRUCTED ENVIRONMENT AND LOCAL SCATTERING

A. A fixed joint host realization

The first calculation asks whether a published joint host realization can consistently supply the gravitational field and

the distributions entering Eqs. (27) and (28). We select the NGC 3201 row with the largest normalized posterior quadrature weight in Eq. (6), retaining all thirteen coordinates. This deterministic selection has weight 0.001764; it is neither a maximum-likelihood estimate nor a posterior mode. A single selected row does not represent the full host posterior or establish a typical cluster environment.

We reconstruct the spherical, nonrotating, multimass lowered-isothermal model associated with the published fit [4, 10, 11]. The observation bundle supplies age 11.5 Gyr and metallicity [Fe/H] = −1.59. Historical stellar-evolution and distribution-function implementations are pinned with their mass-bin and retention conventions. Their exact identity with the original fitting run cannot be established from the released metadata. This is a reproducible compatible reconstruction, not an exact rerun of the original likelihood analysis.

Writing $f_j = m_j F_j$ for a mass-normalized distribution, the reconstructed family has the form

$$f_j(r, \mathbf{u}) = \mathcal{A}_j \exp\left[-\frac{L^2}{2r_{a,j}^2 s_j^2}\right] e^{\mathcal{E}_j} P(g_t, \mathcal{E}_j), \qquad \mathcal{E}_j > 0, \quad (36)$$

and is zero otherwise. Here $L = |\mathbf{r} \times \mathbf{u}|$, $\mathcal{E}_j = [\Psi(r) - u^2/2]/s_j^2$, and P is the regularized lower incomplete gamma function. The normalizations $\mathcal{A}_j$, velocity scales s_j, and species masses come from the same self-consistent solution. The local number distribution used in the collision integrals is explicitly $F_j = f_j/m_j$. The tangential second moment in the numerical host is the sum of two components, so $\beta_j = 1 - \langle u_t^2 \rangle_j / (2\langle u_r^2 \rangle_j)$.

The relative potential is referenced to the distribution's truncation radius r_t. For the isolated spherical continuation with physical potential zero at infinity,

$$\Psi(r) = \Phi_c(r_t) - \Phi_c(r), \quad (37)$$

$$\Phi_c(r) = -\Psi(r) - \frac{GM}{r_t}, \qquad r \le r_t, \quad (38)$$

$$u_{\rm cut}(r) = \sqrt{2\Psi(r)}, \qquad v_{\rm esc,\infty}(r) = \sqrt{2\Psi(r) + 2GM/r_t}. \quad (39)$$

Thus the field-star velocity cutoff is distinct from isolated escape to infinity, and neither is automatically a tidal escape criterion. We preserve the source model's $G = 0.004302\,\mathrm{pc}\,(\mathrm{km\,s^{-1}})^2 M_\odot^{-1}$ throughout its gravity and collision calculation. Its small difference from the catalog-display constant does not enter as an inconsistent force normalization.

The selected realization has $M = 1.7806 \times 10^5\,M_\odot$, $r_h = 10.325\,\mathrm{pc}$, and $r_t = 981.0\,\mathrm{pc}$. Its central density is $2.146 \times 10^3\,M_\odot\,\mathrm{pc}^{-3}$, with $u_{\rm cut}(0) = 17.410\,\mathrm{km\,s^{-1}}$ and $v_{\rm esc,\infty}(0) = 17.455\,\mathrm{km\,s^{-1}}$. The extended formal truncation radius is not adopted as an empirical boundary for NGC 3201. Section VI tests an external-field boundary and an accuracy domain; a physically calibrated outer transport model remains necessary. The separate $34.807\,\mathrm{km\,s^{-1}}$ parameter in the source stellar-evolution prescription is not the present reconstructed escape speed.

TABLE IV. Numerical checks for the fixed NGC 3201 reconstruction. Errors are relative, except the potential error is normalized to its central value. Force checks cover $10^{-3} < r/r_h < 10$; DF checks use four radii. Projected quantities refer to the luminous tracer, not empty remnant tails. These checks establish numerical consistency, not physical calibration.

Check	Maximum error
Integrated mass	3.26×10^{-8}
Half-mass radius	1.20×10^{-6}
Force versus enclosed mass	5.72×10^{-5}
Relative potential	2.35×10^{-7}
Virial balance	9.28×10^{-8}
DF density and raw moments	1.66×10^{-8}
Projected luminous tracer	1.30×10^{-3}
Drift quadrature refinement	5.50×10^{-9}
Diffusion quadrature refinement	1.31×10^{-7}

B. Independent numerical and projected checks

The reconstruction contains sixteen physical mass bins: nine main-sequence bins, five white-dwarf bins, one neutron-star bin, and one black-hole bin. No artificial tracer mass is appended for this observation bundle. We independently integrate density to recover mass and half-mass radius, compare the potential gradient with enclosed mass, and integrate the force to recover the relative potential. The virial check compares $2K + W$ with W, using density-weighted radial and both tangential velocity moments. Positive densities and pressures are interpolated in logarithms to resolve the steep tails of segregated species.

Independent velocity quadrature recovers the density and raw second moments of every species at $r/r_h = 0, 0.1, 1, 3$. Projected luminous-tracer moments are calculated by line-of-sight integration, with the correct radial and tangential projection factors. Table IV reports the numerical errors. The adopted host checks require relative errors below 10^{-3}, or 2×10^{-3} for projected luminous-tracer quantities. These tolerances test implementation accuracy; they are not observational uncertainty intervals.

Figure 4 compares the reconstruction with the released projected posterior envelopes over $0.01 \le R/r_h \le 3$. At the 32 retained source radii, line-of-sight and radial proper-motion dispersions fall within the published central 95.45% envelopes. Tangential proper motion lies inside at 27 radii, and the analogous total-density comparison lies inside at 25. A joint posterior draw need not lie within every pointwise interval; these counts are descriptive, not a goodness-of-fit probability. The comparison neither proves exact historical input compatibility nor replaces an observational refit.

C. Anisotropic coefficient evaluation

We evaluate the weak-scattering kernels without imposing a Maxwellian or a scalar velocity dispersion. To remove the integrable singularity from numerical quadrature, set $\mathbf{v} - \mathbf{u} =$

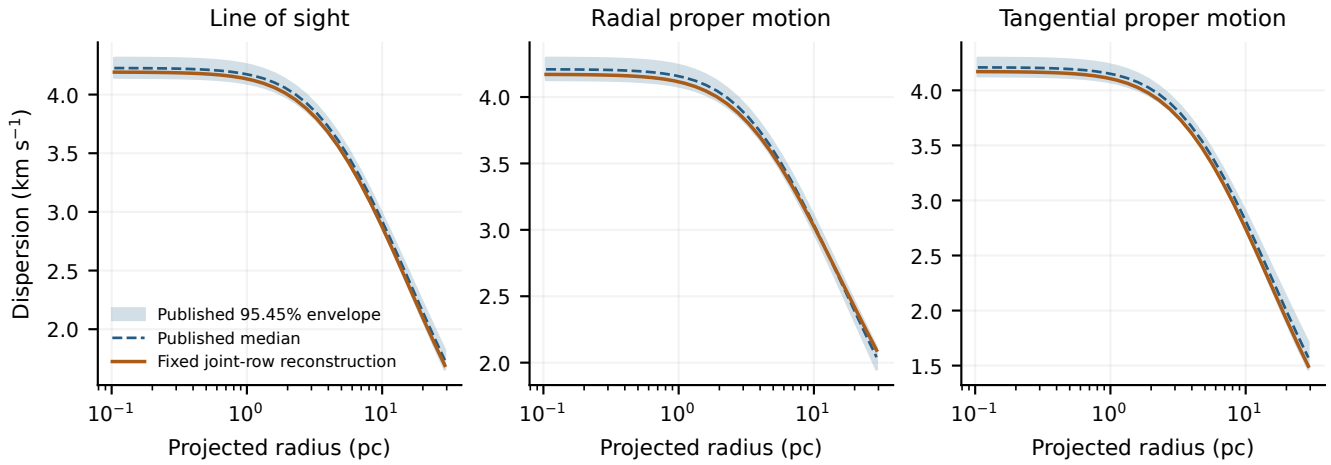


FIG. 4. Projected kinematics of the fixed NGC 3201 reconstruction, compared with published posterior medians and central 95.45% model envelopes. These bands summarize correlated model realizations and are not independent measurements. The selected reconstruction retains one complete posterior row; it does not propagate the host posterior.

$s\mathbf{n}$, with $s \geq 0$ and $|\mathbf{n}| = 1$. Equations (29) and (30) become

$$\partial_i h_j = -\int d\Omega \int ds\, F_j(\mathbf{v} - s\mathbf{n}) n_i, \tag{40}$$

$$\partial_i \partial_k g_j = \int d\Omega \int ds\, s F_j(\mathbf{v} - s\mathbf{n})(\delta_{ik} - n_i n_k). \tag{41}$$

The radial interval is the intersection of the ray with the field-star velocity ball $|\mathbf{u}| < u_{\rm cut}$. Positive quadrature weights preserve the positive-semidefinite diffusion tensor. An independent identity is

$$\operatorname{tr} \mathsf{D} = 8\pi G^2 \sum_j m_j^2 \ln \Lambda_j h_j. \tag{42}$$

Uniform-ball analytic limits, rotations of an anisotropic distribution, this trace identity, and quadrature refinement test the implementation.

The diagnostic remnant mass is $60\,M_\odot$, a controlled probe rather than an inferred merger remnant. We evaluate twenty local states: four radii $r/r_h = 0, 0.1, 1, 3$, zero speed, and radial or tangential velocities at 0.25 and 0.75 of the local stellar cutoff. Results are reported per unit common constant $\ln\Lambda$; no impact-parameter prescription is selected. With distance in pc and speed in km s^{-1}, both raw drift and diffusion require division by 0.97779222 to express their time denominators in Myr. A noise amplitude would instead be obtained by factoring the converted diffusion tensor.

Increasing the angular and ray quadrature orders from 40 to 64 changes drift by at most 5.50×10^{-9} in vector relative norm and diffusion by 1.31×10^{-7} in matrix relative norm on this grid. At zero speed, symmetry requires zero drift and its absolute numerical residual is used. All diffusion tensors are positive semidefinite, and the trace identity holds to 9.0×10^{-15} relative accuracy. This grid validates local evaluation at the tested states, not interpolation along arbitrary recoil trajectories.

TABLE V. Local weak-scattering coefficients for a $60\,M_\odot$ diagnostic remnant with radial speed $0.25\sqrt{2\Psi(r)}$. Both drift magnitude $a_\parallel = -\mathbf{A}\cdot\widehat{\mathbf{v}}$ and diffusion trace are divided by a common constant $\ln\Lambda$. Units are $(\mathrm{km\,s^{-1}})/\mathrm{Myr}$ and $(\mathrm{km\,s^{-1}})^2/\mathrm{Myr}$, respectively. The BH fraction is its contribution to the diffusion trace in the same host. No physical impact-parameter cutoff is selected.

r/r_h	$a_\parallel/\ln\Lambda$	$\mathrm{tr}\mathsf{D}/\ln\Lambda$	BH fraction
0	8.31×10^{-1}	1.04×10^{0}	8.95×10^{-1}
0.1	1.24×10^{-1}	4.87×10^{-2}	3.20×10^{-4}
1	7.13×10^{-4}	2.35×10^{-4}	4.19×10^{-35}
3	4.27×10^{-5}	8.77×10^{-6}	5.40×10^{-46}

D. Species contributions and the limit of smooth diffusion

The selected host contains an expected 4.28 black holes, with total mass $61.2\,M_\odot$ in a bin of individual mass $14.31\,M_\odot$. These are model expectations, not observed object counts. At the center this species supplies approximately 27.7% of mass density but 95.8% of the formal diffusion trace at zero test velocity. At $0.1r_h$, its diffusion fraction has fallen to 8.46×10^{-4}. At nonzero radial speed the decomposition remains strongly position dependent (Table V). Figure 5 compares density and diffusion contributions within this same host; it is not a counterfactual fit obtained by removing black holes.

The enhancement follows from the mass weighting and velocity dependence of Eq. (28), not merely the total density. It also exposes a physical limitation: numerical convergence of a smooth collision integral does not establish that a few heavy objects act as numerous independent weak scatterers. Resolved encounters or an independently calibrated discrete-transition description must test that contribution before it is represented as Gaussian noise. The outer truncation is tested against an external field in Sec. VI. We therefore report local coefficients conditional on this realization and closure, without interpreting them as calibrated damping times, remnant

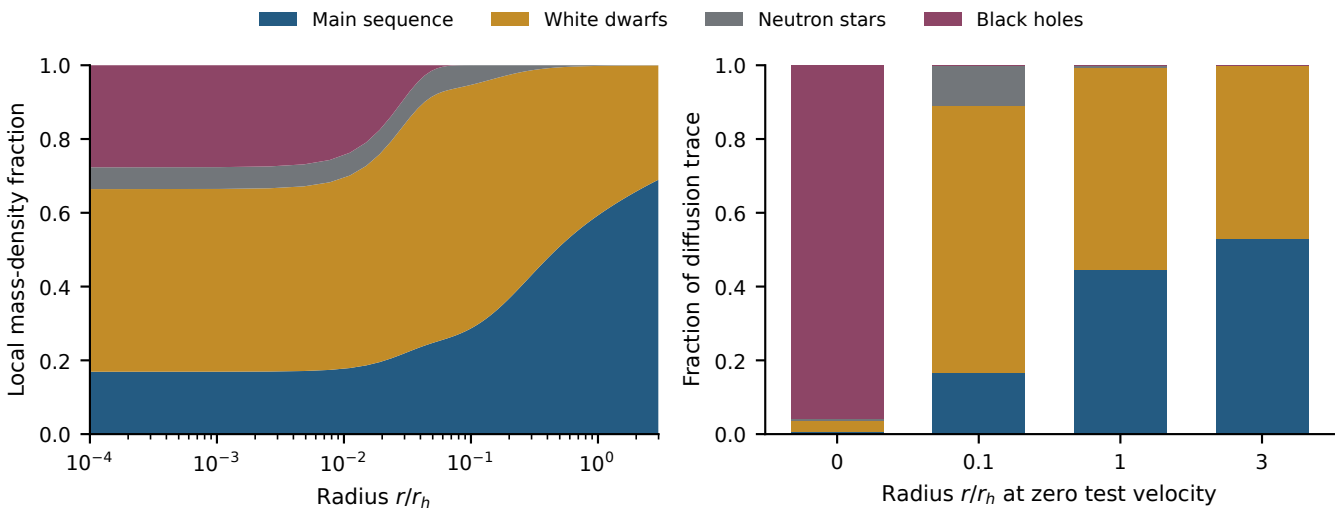


FIG. 5. Mass-density fractions in the reconstructed host (left) and species contributions to the formal diffusion trace at zero test velocity (right). The radial density curves describe the continuous model; the diffusion bars are four evaluated states. The black-hole component has an expected total count of only 4.28, so its dominant central diffusion contribution identifies a continuum-validity question rather than a measured relaxation rate.

paths, or residence outcomes.

The reconstruction does supply the same species densities required for Step 2. It makes the next physical question specific: whether a justified treatment of sparse remnant encounters changes the occupation of dense regions relative to the smooth approximation, with host, launch, cutoff and boundary choices held fixed. Testing that question requires trajectories and their stopping records; the coefficient decomposition alone does not answer it.

VI. OUTER VALIDITY AND TIDAL OPENINGS

A. Boundary definitions and observational scales

The outer boundary constrains the interpretation of the same host used for local scattering. A density-profile limit, a tidal opening, and a stop imposed by an approximation are distinct. Empirical cluster profiles supply fitted limiting radii [12]; tidal dynamics adds a dependence on the Galactic field and the cluster orbit [13]. Even above the critical energy of a stationary tidal problem, escape can be delayed [14]. A crossing cannot therefore be converted directly into a merger-retention probability.

For NGC 3201, Wan et al. [15] distinguish a King limiting radius of 36.1 pc from a cited Jacobi estimate of 83.46 pc. Our captured structural catalog lists 85.86 pc, with its definition referring to Eq. (8) of Webb et al. [16]: an eccentricity- and phase-dependent correction to a pericentric limiting scale. We do not invert that value to infer an instantaneous tidal tensor or average these radii as independent observations.

B. A controlled external-field calculation

We keep the reconstructed spherical mass distribution fixed and prescribe

$$\Phi_G(\mathbf{R}) = v_c^2 \ln(|\mathbf{R}|/R_0), \tag{43}$$

where R_0 sets an irrelevant potential constant. The baseline uses the catalog Galactocentric distance $R = 8.93\,\mathrm{kpc}$ and a flat-rotation benchmark $v_c = 220\,\mathrm{km\,s^{-1}}$. The latter is also used in the illustrative circular comparison of Ref. [15]. This is a specified environmental test, not a fitted Milky Way potential or a reconstruction of NGC 3201's eccentric orbit. The external distance is held separately from the selected host's inferred heliocentric distance; their joint uncertainty is not propagated.

In the translating nonrotating frame of Eq. (14), the linear tidal tensor is $\mathsf{T} = -\nabla\nabla\Phi_G$. Let $\widehat{\mathbf{R}}$ be the Galactic radial direction. For Eq. (43),

$$\mathsf{T} = \frac{v_c^2}{R^2}\left(2\widehat{\mathbf{R}}\widehat{\mathbf{R}}^{\mathsf{T}} - \mathsf{I}\right). \tag{44}$$

For a circular orbit, a frame rotating at $\Omega = v_c/R$ additionally has centrifugal acceleration. In axes along the Galactic radial, orbital tangential, and orbital normal directions, its effective tensor is

$$\mathsf{T}_{\mathrm{eff}} = \frac{v_c^2}{R^2}\,\mathrm{diag}(2, 0, -1). \tag{45}$$

The velocity-dependent Coriolis term still belongs to rotating-frame dynamics. It does not alter a stationary saddle, and neither it nor the centrifugal tensor is inserted into the nonrotating equation.

Balancing cluster gravity with the extensive effective direction gives the circular Hill estimate for an extended spherical host,

$$\frac{GM(<r_J)}{r_J^3} = \lambda_1, \qquad \lambda_1 = \frac{2v_c^2}{R^2}. \tag{46}$$

The point-mass result $(GM/\lambda_1)^{1/3}$ follows when all cluster mass lies inside the saddle. Here we retain the reconstructed enclosed mass. As an independent check, the exact circular inward and outward saddle distances satisfy

$$\frac{GM(<r_\pm)}{r_\pm^3} = \frac{v_c^2}{R(R \pm r_\pm)} + \frac{v_c^2}{R^2}, \tag{47}$$

with the minus sign toward the Galactic center. These conditions compare the tidal expansion with exact differential Galactic and centrifugal accelerations without integrating a trajectory.

C. An accuracy boundary for an isolated control

A tidal opening does not specify where neglecting the Galaxy becomes inaccurate. In the same logarithmic field, the exact maximum differential acceleration at fixed $0 < r < R$ is attained toward the Galactic center:

$$\max_{|\mathbf{r}|=r} |\mathbf{a}_{\rm ext}| = \frac{v_c^2 r}{R(R-r)}. \tag{48}$$

Dividing by the spherical cluster acceleration defines

$$\eta(r) = \frac{v_c^2 r^3}{GM(<r)R(R-r)}. \tag{49}$$

We propose a conditional accuracy boundary

$$r_\epsilon = \inf\{r > 0 : \eta(r) > \epsilon\}, \tag{50}$$

using $\epsilon = 0.01$ for an isolated-host control and comparing 0.03 and 0.1. The evaluated host has increasing η throughout the relevant interval, so its limiting radius is obtained from $\eta(r_\epsilon) = \epsilon$. This tolerance is an analysis choice, not a physical law. It bounds an instantaneous omitted force under the prescribed field; it does not bound accumulated orbit error, establish a tidally consistent equilibrium, or justify the collision closure.

D. Findings and implications for transport

The baseline gives $r_J = 84.13$ pc, compared with 85.77 pc for the point-mass approximation. Exact circular saddles are at 84.00 and 84.27 pc. Agreement in scale with the published 83.46 pc estimate does not independently validate the host or its velocity distribution. The radius is conditional on the adopted Galactic field and one host realization.

The force limits are $r_{1\%} = 20.67$ pc, $r_{3\%} = 31.12$ pc, and $r_{10\%} = 47.94$ pc. At the existing coefficient radii r_h and $3r_h$, the maximum omitted force is 0.175% and 2.962% of cluster gravity, respectively. Thus the outermost local kernel remains a defined integral of its specified distribution, but its use in an isolated trajectory exceeds the proposed 1% force budget. Table VI varies v_c by ±10% and includes a separate 13 kpc circular surrogate discussed in Ref. [15]. These scenarios are assumption tests, not an orbit track or confidence intervals.

TABLE VI. Conditional boundary calculations in a spherical logarithmic Galactic field, keeping the NGC 3201 host fixed. r_J uses enclosed mass. $r_{1\%}$ bounds the maximum neglected Galactic acceleration by 1% of the cluster acceleration. Cases are sensitivity settings, not uncertainty intervals; the 13 kpc case is a circular surrogate.

R (kpc)	v_c (km/s)	r_J (pc)	$r_{1\%}$ (pc)	$\eta(3r_h)$
8.93	220	84.13	20.67	2.96%
8.93	198	90.44	22.39	2.40%
8.93	242	78.79	19.22	3.58%
13.00	220	108.79	27.40	1.40%

The unmodified host places 5.63% of its mass beyond the baseline saddle, while its formal truncation is 981 pc. This is a consistency warning about the outer distribution, not a predicted escaped fraction. Simply removing this material would not yield a self-consistent tidal distribution function. An exterior boundary requires a host appropriate to the imposed tide as well as a remnant event rule.

In the conservative circular Hill problem, the critical specific energy is $E_{J,c} = \Phi_{\rm cl}(r_J) - \lambda_1 r_J^2/2$. The corresponding central opening speed,

$$v_{\rm open}(0) = \sqrt{2[E_{J,c} - \Phi_{\rm cl}(0)]}, \tag{51}$$

is $16.68\,{\rm km\,s^{-1}}$. It refers to the remnant's full rotating-frame speed at the center, not its recoil alone. It is neither a measured escape speed nor a retention threshold with friction and diffusion. Noncircular tides and collisional changes remove the conservation on which this energy argument rests.

Boundary roots have relative residuals below 9.5×10^{-13}. Independent density integration agrees with the enclosed mass at the tested boundaries to 1.2×10^{-7}; thinning the radial mesh changes inferred radii by less than 1.5×10^{-7}. Analytic point-mass and Plummer limits, the exact angular force maximum, and force–potential consistency give additional checks. These errors describe numerics, not Galactic or host uncertainty.

For an isolated control launched strictly inside $r_{1\%}$, its first outward crossing is proposed as $\tau_{\rm valid}$ in Eq. (33). A launch already outside has $\tau_{\rm valid} = 0$; a boundary launch needs a declared side convention. This stop would censor the trajectory and accumulated exposure. It would not be labeled escape or permanent loss of access to the center. A trajectory including the external force has a different validity criterion. Extending the calculation to the tidal outskirts requires a time-dependent Galactic trajectory, an appropriate outer host, and explicit records of departures and possible returns. The sparse-remnant issue identified in Sec. V remains a separate prerequisite for physical collisional transport.

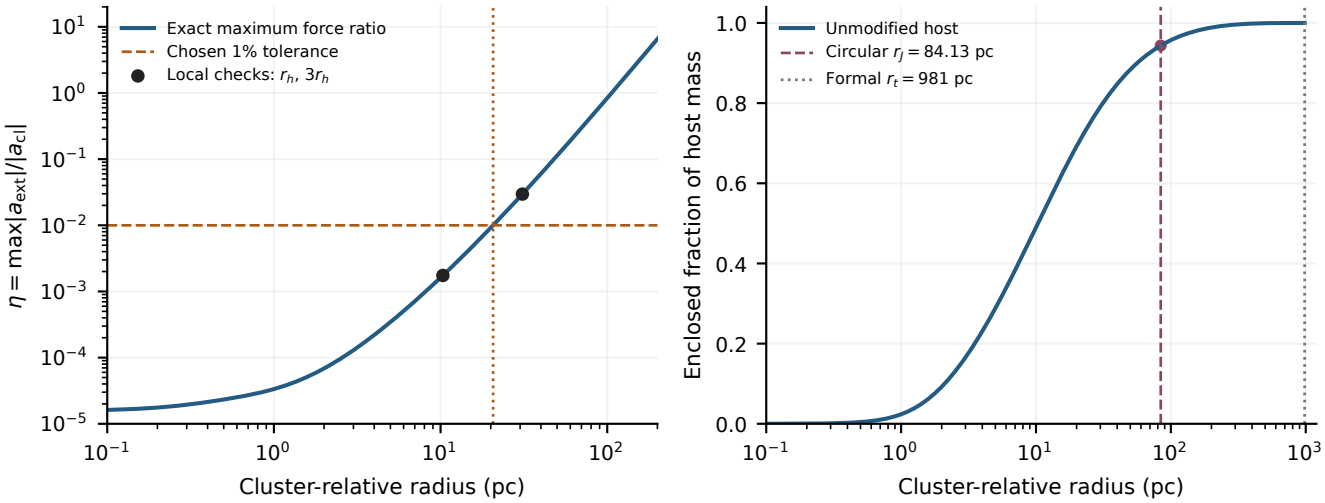


FIG. 6. Baseline boundary diagnostics for the fixed NGC 3201 realization. Left: the maximum differential Galactic force relative to cluster gravity, with the chosen 1% tolerance and two existing local-coefficient evaluation radii. Right: enclosed mass in the unmodified host, compared with the circular extended-host saddle and formal distribution-function truncation. Exterior mass is not classified as escaped by this calculation.

VII. GALACTIC ORBIT AND TIME-DEPENDENT HOST SUPPORT

A. Observed phase point and defined reference orbits

The static boundary calculation does not specify how the tide changes along NGC 3201's orbit. We therefore integrate the cluster's Galactic reference motion and apply the resulting field to the same host as a diagnostic. This calculation evolves neither a kicked remnant nor the cluster's internal distribution.

The database's orbital table supplies a Galactocentric Cartesian point

$$\mathbf{R}_0 = (7.59, -4.65, 0.71)\,\mathrm{kpc},$$
$$\mathbf{V}_0 = (-250.21, -200.78, 149.10)\,\mathrm{km\,s^{-1}}. \qquad (52)$$

Its axes point toward the Sun, along solar motion, and north of the Galactic plane. We preserve the published point and its rounding. The table also provides heliocentric distance $D = 4.74 \pm 0.04$ kpc, proper motions $(\mu_{\alpha*}, \mu_\delta) = (8.350, -1.976)\,\mathrm{mas\,yr^{-1}}$, and radial velocity $495.38 \pm 0.06\,\mathrm{km\,s^{-1}}$. Both proper-motion errors are $0.005\,\mathrm{mas\,yr^{-1}}$, with correlation 0.18; a full six-dimensional covariance is not supplied. These compiled constraints retain the distance and astrometric sources in Refs. [7, 9].[2]

We adopt Model I of Irrgang et al. [17], a fixed axisymmetric bulge–disk–halo potential, as a reproducible reference used by the catalog's modeling framework. Its Table 1 parameters and exact potential unit $100\,\mathrm{km^2\,s^{-2}}$ are retained. This is not a uniquely inferred Galactic potential. No bar, spiral structure, secular evolution, or Galactic-parameter posterior is added.

Distance changes must alter transverse velocity as well as position. With line-of-sight unit vector $\mathbf{n}$ and proper-motion tangent vectors $\mathbf{e}_\alpha, \mathbf{e}_\delta$ in the same axes,

$$\Delta\mathbf{R} = \Delta D\,\mathbf{n},$$
$$\Delta\mathbf{V} = k_\mu \Delta D(\mu_{\alpha*}\mathbf{e}_\alpha + \mu_\delta \mathbf{e}_\delta), \qquad (53)$$

where $k_\mu = 4.74047046$ in the units of Eq. (2). We compare the catalog distance, its quoted error shifts, and the selected host's 4.54335 kpc distance. The latter retains that host's distance coordinate but is not a joint astrometric–host posterior. The catalog Cartesian anchor preserves its solar-frame transformation; missing cross-covariances and solar-frame uncertainties are not invented.

Each reference orbit obeys $\ddot{\mathbf{R}} = -\nabla\Phi_G$ for 1 Gyr forward in the specified potential. Spherical radial turning points and Galactic-plane crossings are detected as events. This horizon samples repeated phases; it is not a forecast that the internal host remains unchanged for a gigayear. Energy and axial angular momentum are conserved in the adopted Galactic model. Their relative numerical drifts remain below 10^{-8}, and refined integration and time reversal meet the declared position and velocity tolerances.

B. A sufficient isolated-force domain along the orbit

Write $\mathsf{T}(\mathbf{R}) = -\nabla\nabla\Phi_G(\mathbf{R})$. For a bound $B(\mathbf{R}, r)$ satisfying

$$\|\mathsf{T}(\mathbf{R}+\mathbf{s})\|_2 \le B(\mathbf{R}, r) \quad \text{for } |\mathbf{s}| \le r, \qquad (54)$$

the exact differential Galactic force obeys

$$|\mathbf{a}_{\mathrm{ext}}| \le rB(\mathbf{R}, r). \qquad (55)$$

[2] The orbital input uses the database's /newdata/ orbital table, without merging it with the root-path version.

This follows by integrating the force derivative along the segment from $\mathbf{R}$ to $\mathbf{R}+\mathbf{r}$. Appendix A supplies an analytic componentwise bound for the adopted potential. It is valid for all directions in the ball and does not infer a supremum from finitely sampled angles.

A sufficient radius for a chosen omitted-force tolerance solves

$$\eta_B(r,t) = \frac{B(\mathbf{R}(t),r)r^3}{GM(<r)} = \epsilon, \qquad \epsilon = 0.01. \tag{56}$$

This generalizes the static calculation without identifying the radius with escape. It controls an instantaneous neglected acceleration under the supplied host; it is neither a cumulative trajectory-error guarantee nor evidence that the stellar distribution is tidally self-consistent.

C. Frozen-frame scales and forcing timescales

For comparison, define a frame whose radial direction follows $\mathbf{R}$ with minimal instantaneous rotation,

$$\boldsymbol{\omega} = \frac{\mathbf{R}\times\mathbf{V}}{|\mathbf{R}|^2},$$
$$\mathsf{T}_f = \mathsf{T} + |\boldsymbol{\omega}|^2\mathsf{I} - \boldsymbol{\omega}\boldsymbol{\omega}^{\mathsf{T}}. \tag{57}$$

We report r_f from $GM(<r_f)/r_f^3 = \lambda_{\max}(\mathsf{T}_f)$. On this inclined eccentric orbit, r_f is only a frozen-frame diagnostic. The Euler acceleration $-\dot{\boldsymbol{\omega}}\times\mathbf{r}$ is omitted from that definition, and a stationary Jacobi surface does not follow. Actual rotating-frame dynamics would also require Coriolis acceleration. The transport formulation in nonrotating axes continues to use the exact differential Galactic force.

An additional indicator compares internal dynamical time to changes in forcing:

$$\mathcal{A}(r,t) = \sqrt{\frac{r^3}{GM(<r)}}\frac{\|\dot{\mathsf{T}}\|_2}{\|\mathsf{T}\|_2}. \tag{58}$$

Time units are converted consistently. This quantity includes tensor orientation changes in the fixed axes. We separately evaluate variation of the tensor norm alone. Neither indicator is an inferred heating rate or a calibrated adiabatic correction.

D. Findings and limits of the frozen host

The catalog-reference orbit has mean pericenter 8.333 kpc and apocenter 25.362 kpc, compared with catalog estimates 8.33 and 25.82 kpc. The apocenter difference is 1.77%. This is a descriptive scale check; the original averaging procedure and complete model configuration are not reconstructed.

For the selected-host-distance orbit, mean pericenter and apocenter are 8.215 and 23.810 kpc. The sufficient 1% force radius varies from approximately 8.2 to 39.8 pc. Halving output spacing changes its minimum by 0.0674%, giving 8.173 pc; rounding downward to a whole parsec motivates an 8 pc fixed isolated-control domain for this specific calculation. It does not transfer unchanged to another host or Galactic model. Initial points outside a control domain are invalid at time zero; interior trajectories are censored on validity exit, without being labeled escaped.

The bounds on neglected force reach 1.73% at r_h and 29.5% at $3r_h$. The static logarithmic-field limit of 20.67 pc therefore cannot be carried unchanged through the orbit. A calculation retaining exact Galactic forcing does not require this omitted-force stop, but still requires an adequate host and collision model.

The frozen-frame scale spans 62.1–183.9 pc, and the unmodified host puts 1.37–8.74% of its mass beyond it. These are neither stripped-mass estimates nor successive equilibria. The ratio $|\dot{\omega}|/\lambda_{\max}(\mathsf{T}_f)$ reaches 0.67, reinforcing the distinction from a stationary tidal-opening calculation. At r_h, $\mathcal{A}$ reaches 1.16; norm variation alone gives 1.14. Thus rapid forcing is not only a tensor-orientation effect. The indicators identify phases where a slowly adjusting host approximation needs testing rather than supplying that test's outcome.

Self-consistent nonspherical equilibria can be constructed for circular orbits in a frozen tidal field [18]; that construction does not establish equilibrium under an eccentric orbit's changing tide. A consistent evolving host instead requires species equations of the form

$$\partial_t F_j + \mathbf{u}\cdot\nabla_{\mathbf{r}}F_j + \mathbf{a}_g\cdot\nabla_{\mathbf{u}}F_j = C_j[\{F_k\}], \tag{59}$$

coupled to Eq. (11), appropriate boundaries, and specified collisional physics. We have not solved this host-evolution problem. The orbit supplies its external forcing and tests the domain of the fixed-host approximation; it does not repair the outer distribution by deleting exterior stars. The central sparse-remnant scattering issue remains independent of this environmental calculation.

VIII. STEP 2: RESIDENCE AND ENVIRONMENTAL EXPOSURE

A. Finite-region entry and re-entry

Specify an operational region by a level-set function,

$$C(t) = \{\mathbf{r} : g(\mathbf{r},t) < 0\}. \tag{60}$$

The function is negative inside, positive outside, and zero on the boundary. This g describes geometry and is distinct from the velocity-space potentials g_j in Eq. (26). A sphere may be represented by $g = |\mathbf{r}| - r_c(t)$, but neither a shape nor a radius is selected. A projected observational core radius is not automatically the required spatial boundary.

For an initially interior launch, define first exit and subsequent re-entry as

$$\tau_{\text{out}} = \inf\{t > 0 : g(\mathbf{r}_t,t) > 0\}, \tag{61}$$
$$\tau_{\text{re}} = \inf\{t > \tau_{\text{out}} : g(\mathbf{r}_t,t) < 0\}. \tag{62}$$

The strict opposite-side conditions exclude an isolated tangential contact. For a continuous path, the infimum identifies the

TABLE VII. Galactic reference-orbit diagnostics over the defined 1 Gyr horizon in Irrgang Model I. The host mass profile is fixed. D is heliocentric distance; mean pericenter and apocenter are spherical Galactocentric radii from detected turning events. The intervals show orbit-phase variation of the sufficient 1% isolated-force radius r_ϵ and frozen-frame diagnostic r_f, not posterior uncertainties.

Case	D (kpc)	$\langle R_p \rangle$ (kpc)	$\langle R_a \rangle$ (kpc)	r_ϵ (pc)	r_f (pc)
Catalog distance	4.740	8.33	25.36	9.1–41.8	63.0–193.6
Selected-host distance	4.543	8.22	23.81	8.2–39.8	62.1–183.9
Catalog distance minus quoted error	4.700	8.31	25.03	9.0–41.4	63.0–191.6
Catalog distance plus quoted error	4.780	8.36	25.70	9.1–42.3	63.1–195.7

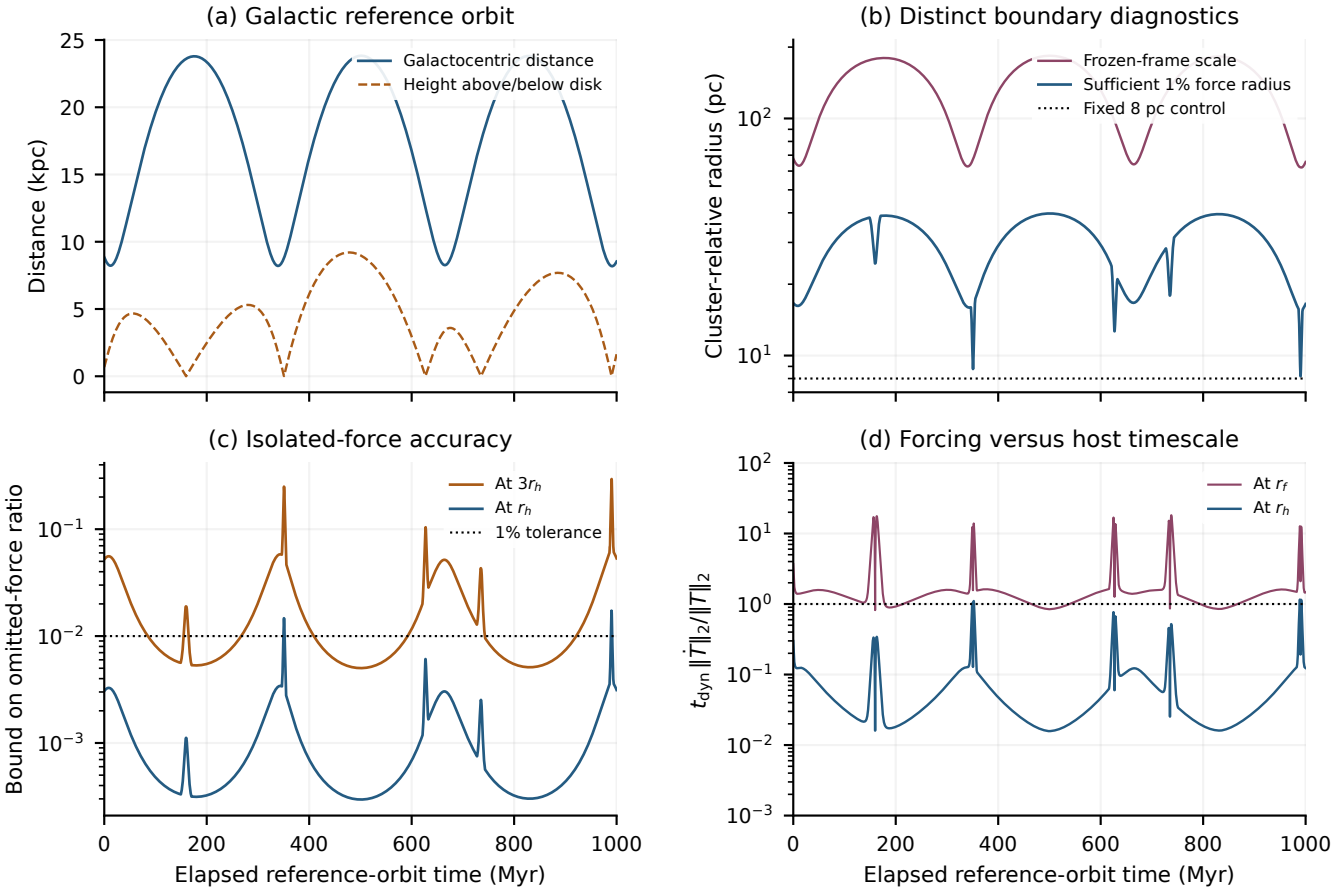


FIG. 7. The reference orbit using the selected host's distance, with the internal host held fixed only for the diagnostic. (a) Galactic radius and absolute height. (b) Sufficient isolated-force radius and frozen-frame scale; neither is a measured cluster edge. (c) Bounds on the omitted Galactic force at the existing local-coefficient radii. (d) Changes in forcing relative to host dynamical time. The proposed fixed 8 pc control is conditional on this orbit and host, and does not classify escape.

boundary time associated with a sign-changing crossing. An observed re-entry delay is $\tau_{\rm re} - \tau_{\rm out}$.

For an initially exterior launch, define first entry by

$$\tau_{\rm ent} = \inf\{t > 0 : g(\mathbf{r}_t, t) < 0\}. \tag{63}$$

A boundary launch requires a declared initial-side convention. Classification may use the first definite side reached, while recording the contact separately. It must not create an artificial return event. A path that never exits has no observed re-entry, and the convention $\inf \varnothing = +\infty$ does not establish that a censored event is physically impossible.

For a differentiable boundary,

$$\frac{d}{dt} g(\mathbf{r}_t, t) = \partial_t g + \mathbf{v}_t \cdot \nabla g. \tag{64}$$

Boundary evolution can change membership without inward remnant motion. Both the crossing and its origin must remain identifiable. Sign changes remain the definition even when the derivative vanishes. Separate inner and outer event surfaces can control numerical recrossings only if their effects are reported; they do not silently redefine $C(t)$.

Only events observed before the modeled endpoint are recorded as completed. Simultaneous crossing and stopping

events require an ordering convention. Return flags and termination causes are separate fields: a path can re-enter the region and later escape.

B. Path-integrated observables

The residence accumulated before termination is

$$C_T = \int_0^{T_*} \mathbf{1}_{\{\mathbf{r}_t \in \mathcal{C}(t)\}}\, dt. \tag{65}$$

It has units of time and obeys $0 \le C_T \le T_* \le T$. The ratio C_T/T uses a common horizon across realizations. If a description stops before T because subsequent motion is unresolved, this ratio is not a complete-horizon physical residence fraction. The alternative C_T/T_*, for $T_* > 0$, describes the modeled interval and answers a different question.

For each species, define mass-density exposure

$$J_{j,T} = \int_0^{T_*} \rho_j(\mathbf{r}_t, t)\, dt. \tag{66}$$

Its dimensions are mass times inverse volume times time. A normalized exposure additionally requires a declared positive reference:

$$I_{j,T} = \int_0^{T_*} \frac{\rho_j(\mathbf{r}_t, t)}{\rho_{\mathrm{ref},j}(t)}\, dt, \tag{67}$$
$$\rho_{\mathrm{ref},j}(t) > 0.$$

It has units of time. The reference is a convention, not an automatically inferred central density. Unless it bounds the sampled density, $I_{j,T}/T$ need not lie between zero and one.

Density may vary even while a trajectory remains inside $\mathcal{C}(t)$; density outside that region may also be nonzero. Region-restricted exposure can be defined by inserting the same indicator into Eq. (66), but it is a distinct observable. None of these integrals is an encounter count.

C. Time-augmented occupation

Preserve the chronological relation between motion and environment through

$$\mu_T(\mathcal{A}) = \int_0^{T_*} \mathbf{1}_{\{(t,\mathbf{r}_t,\mathbf{v}_t)\in\mathcal{A}\}}\, dt, \tag{68}$$

where $\mathcal{A}$ is a measurable subset of $[0,T]\times\mathbb{R}^3\times\mathbb{R}^3$. The measure has total mass T_* and units of time. Equations (65) and (66) are weighted integrals of that same measure:

$$C_T = \int \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, \mu_T(dt, d^3r, d^3v), \tag{69}$$

$$J_{j,T} = \int \rho_j(\mathbf{r}, t)\, \mu_T(dt, d^3r, d^3v). \tag{70}$$

Removing time labels can destroy the association with an evolving host; removing velocity can destroy information needed for a later dynamical kernel. Aggregated exposure therefore accompanies the trajectory and event record rather than replacing them.

D. Ensemble averages and incomplete histories

At fixed Q, nonnegative path integrands give the identities

$$\mathbb{E}[C_T \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \mathbf{1}_{\{g(\mathbf{r},t)<0\}}\, p_{\mathrm{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q), \tag{71}$$

$$\mathbb{E}[J_{j,T} \mid Q] = \int_0^T dt \int d^3r\, d^3v \times \rho_j(\mathbf{r}, t)\, p_{\mathrm{alive}}(\mathbf{r}, \mathbf{v}, t \mid Q). \tag{72}$$

If a density conditional on survival is used instead, it must be multiplied by $S(t \mid Q)$. Renormalizing the remaining trajectories silently changes the population being averaged. Equations (71) and (72) provide a future consistency comparison between trajectory accumulation and a transport-density solution.

The event record must distinguish no exit observed, entry or re-entry observed, finite-horizon censoring, declared escape, unresolved strong interactions, and loss of model validity. Escape can compete with return; it is not interchangeable with numerical failure. A spatial boundary crossing counts as escape only under the declared escape rule.

After an unresolved stop, complete-horizon residence and exposure remain unknown. The nonnegative accumulated integrals describe the modeled interval, while extrapolation needs a continuation model. Censoring can depend on the path and environment. Treating it as independent would be an additional statistical assumption requiring justification.

IX. CONDITIONING AND CALCULATION REQUIREMENTS

A. Physical and inferential uncertainty

Host uncertainty, launch uncertainty, and physical scattering are different sources of variation. For a trajectory observable O, their joint accounting takes the form

$$\mathbb{E}[O \mid \mathcal{D}, \mathcal{M}] = \int d\Theta\, p(\Theta \mid \mathcal{D}, \mathcal{M}) \times \int dz_0\, \pi_0(z_0 \mid \Theta, \mathcal{D}, \mathcal{M}) \times \mathbb{E}_{\mathrm{path}}[O \mid \Theta, z_0, \mathcal{D}, \mathcal{M}]. \tag{73}$$

The inner expectation includes the Wiener forcing and any additional randomness in a supplied stopping-event model. This factorization preserves conditional dependence. Merger selection can change host weights as well as launch properties. Alternative host families require separate conditioning or explicit model probabilities. Region definitions, reference densities, and any selection-dependent horizon must also be specified consistently.

Observed stellar kinematics constrain a host through an inference problem; they do not directly provide the entire potential, hidden remnant populations, or collision coefficients. The

posterior summaries in Sec. III quantify dependence within published candidate fits. They do not determine probabilities for competing host families or their merger-conditioned weights.

For a fixed observed cluster c, a compatible reconstructed host law $\Pi_c(d\Theta \mid \mathcal{D}_c)$ would enter a trajectory-observable distribution as

$$\Pr_c(O \in A \mid \mathcal{D}_c, \lambda) = \int \Pr(O \in A \mid \Theta, \lambda)\, \Pi_c(d\Theta \mid \mathcal{D}_c). \quad (74)$$

Here A is a measurable set of observable values, not the drift vector $\mathbf{A}$, and λ retains the declared launch, recoil, collision, boundary, and horizon choices. The integrand includes the launch and path averaging in Eq. (73). This expression is a conditional modeling prescription, not an evaluated prediction. A merger-selected population additionally needs the host reweighting in Eq. (73); the six clusters are not assigned universal population weights.

Each host realization must supply the same F_j to Eqs. (9)–(13) and (25)–(28). The density used for exposure must come from that realization as well. Section V constructs those local functions for one fixed realization and checks its projected predictions. Extension to the joint posterior and competing host prescriptions remains necessary, and compatibility with projected kinematics does not validate a collisional closure. Present-day observations also do not identify the stellar environment at a historical merger epoch.

B. Information passed between the two steps

A calculation becomes defined only when its physical inputs and event conventions are supplied. These include the units and frame; species masses and distributions; internal and external gravity with boundary conditions; the joint launch law; collision closure and scale cutoffs; physical horizon; region geometry; and escape, strong-event, and validity criteria. An unspecified input is not a zero-valued effect.

Step 1 must preserve physical times, positions, velocities, remnant-mass labels, host and launch identifiers, original realization weights, and termination causes. A density-based calculation must preserve its conditioning and absorbed or killed probability. Reproducibility requires the source of each input and enough numerical information to reconstruct coefficient evaluation, stochastic realizations, and error control.

Step 2 attaches crossing events, residence, species exposures, and a time-augmented occupation measure. Every summary retains its denominator convention, modeled endpoint, and status. A path that becomes unresolved is not filled by an assumed central rest state. The same host and epoch must supply the density used for exposure and the distribution used to drive transport.

C. Acceptance criteria before physical interpretation

The host and local-coefficient checks in Sec. V establish dimensional consistency, force–potential agreement, number-to-mass normalization, and positive-semidefinite diffusion for the stated realization and evaluation grid. The following trajectory requirements remain unexecuted. The implementation must preserve the Itô convention and account for probability lost at each boundary or stop. Appropriate conservative and analytically justified stochastic limits provide independent checks, including the energy identity in Eq. (32).

Entry and residence calculations must distinguish tangencies from crossings, handle interior, exterior, and boundary launches, and account for moving regions. Brief passages must remain resolved under a declared refinement procedure. Convergence is needed for the relevant distributions and accumulated observables, not solely for a few representative trajectories. Numerical and sampling errors must be separated from uncertainty in physical inputs.

Where both representations exist, path averages and the survival-weighted identities in Eqs. (71) and (72) should agree within declared numerical accuracy. Comparisons testing a physical simplification must hold the conditional population, horizon, region, and remaining inputs fixed. A useful result could identify a regime in which a reduction preserves the observables, or a discrepancy establishing which additional information is required.

Internal agreement does not calibrate the collision model. Comparison with independently resolved dynamics appropriate to the host and encounter regime is needed before interpreting predictions as quantitatively reliable. Host and coefficient quadrature controls are specified for the completed calculation. A trajectory solver, ensemble precision, physical encounter-scale partition and calibration test have not yet been supplied.

X. DISCUSSION

The central object of the formulation is a merger-conditioned, time-resolved distribution of remnant motion. Finite-region entry, residence, and exposure are functionals of that distribution. Keeping their definitions separate makes explicit why remaining bound, returning to a region, and spending time in a dense environment are different physical questions.

The description permits off-center launches, inherited progenitor motion, anisotropic and streaming stellar populations, time-dependent gravity, and stochastic velocity changes. These dependences are available in the equations; they are not all measured or independently adjustable parameters. A physical realization must justify how they are constrained together and where the adopted closure applies.

The observational analysis makes several requirements concrete. Common units allow direct display comparisons, but source overlap and differing stellar selections prevent automatic construction of a shared likelihood. Strong mass–radius dependence in some published posteriors requires complete host realizations rather than independent marginal draws. Differences between structural estimates make the host prescription itself an uncertainty to examine. The subsequent reconstruction supplies one candidate local scattering distribution,

while its sparse heavy-remnant component illustrates why a consistent distribution still requires a physical validity test.

For a controlled comparison, define

$$\widetilde{C} = \frac{C_T}{t_0}, \qquad \widetilde{J}_j = \frac{J_{j,T} r_h^3}{M t_0}. \tag{75}$$

A testable next hypothesis is that host structure and stellar-mass composition leave differences in these dimensionless observables after fixing the dimensionless launch law, remnant-to-host mass ratio, T/t_0, region scaled by r_h, external-field treatment, collision prescription and termination conventions. The current analysis motivates this hypothesis by identifying structural variation and correlated host uncertainty; it does not test the transport consequence. A controlled comparison must extend the fixed-host reconstruction to compatible joint realizations, justify the treatment of sparse encounters, and compare the same dimensionless launch and event definitions while propagating host uncertainty. The gravitational scales alone cannot decide whether the resulting trajectory observables coincide.

The static and orbit-dependent boundary calculations show why a single formal truncation radius is insufficient: the selected external field gives distinct approximation and tidal-opening scales, and the outer host remains to be made consistent with that field. The observed-phase-point orbit supplies explicit time dependence, while disk passages restrict the isolated-control domain and expose rapid forcing that a host-evolution calculation must address.

The formulation supplies the conditional evolution and accumulation problem needed for that comparison. Continued motion after entry, survival weighting, and termination status remain explicit. Its connection to observed clusters is supported by measured coverage, published host distributions, a numerically checked reconstruction and local anisotropic collision kernels. The dominant central contribution of a few modeled black holes identifies the encounter regime that must be resolved before calibrated return and residence predictions follow.

Timed position–velocity histories can eventually be combined with a separately justified companion population and interaction law. Residence and density exposure alone do not supply either ingredient. Companion acquisition, binary evolution, subsequent mergers, and inference from observed gravitational-wave sources therefore require additional physical descriptions beyond the two steps defined here.

Appendix A: Conservative Galactic-force bound

All lengths in this derivation share one unit. Let $s = |\mathbf{R}|$, $\varpi = \sqrt{X^2 + Y^2}$, and ℓ be the radius of the spatial ball used for the force bound. Define

$$\begin{aligned} s_- &= s - \ell, \qquad \varpi_- = \max(\varpi - \ell, 0), \\ \zeta_- &= \sqrt{\max(|Z| - \ell, 0)^2 + b_d^2}, \\ d_- &= \sqrt{\varpi_-^2 + (a_d + \zeta_-)^2}. \end{aligned} \tag{A1}$$

For Irrgang Model I, write the potential amplitudes as $C_b = GM_b$, $C_d = GM_d$, and $C_h = GM_h$, with scale lengths b_b, a_d, b_d, a_h and halo cutoff Λ. If $s_- > 0$ and $s + \ell < \Lambda$, sufficient component bounds are

$$\begin{aligned} B_b &= \frac{2C_b}{(s_-^2 + b_b^2)^{3/2}}, \\ B_d &= \frac{C_d}{d_-^3} \max\left(2, 1 + \frac{a_d b_d^2}{\zeta_-^3}\right), \\ B_h &= \frac{C_h}{a_h s_- (a_h + s_-)}, \qquad B = B_b + B_d + B_h. \end{aligned} \tag{A2}$$

The bulge and $\gamma = 2$ halo expressions bound their radial and tangential Hessian eigenvalues. For the disk, let $\zeta = \sqrt{Z^2 + b_d^2}$, $d^2 = X^2 + Y^2 + (a_d + \zeta)^2$, and

$$\begin{aligned} \mathbf{q} &= (X, Y, (a_d + \zeta)Z/\zeta), \\ \mathsf{J} &= \mathrm{diag}(1, 1, 1 + a_d b_d^2/\zeta^3). \end{aligned} \tag{A3}$$

Its Hessian is $C_d(\mathsf{J}/d^3 - 3\mathbf{q}\mathbf{q}^\mathsf{T}/d^5)$. Since $\mathsf{J} \geq \mathsf{I}$ and $|\mathbf{q}| \leq d$, its eigenvalues lie between $-2C_d/d^3$ and $C_d J_{zz}/d^3$. The conservative spatial minima in Eq. (A1) then imply B_d. Summing operator norms bounds the full tensor, so Eq. (55) applies to the exact force difference throughout the ball, not just its linear tidal approximation.

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